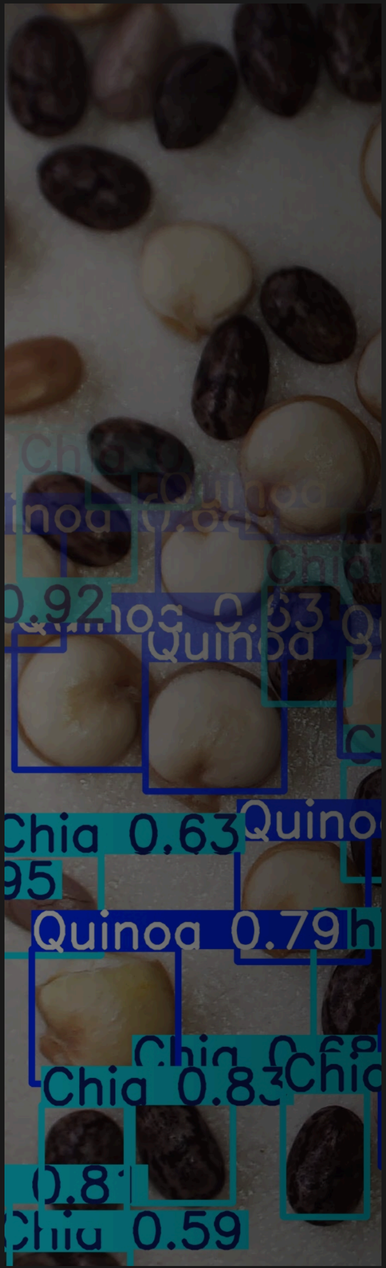


EPFL

Center for Imaging

AMLD

HANDS-ON COMPUTER VISION
ON MICROSCOPES

Workshop Team (*EPFL Center for Imaging*):

Mallory Wittwer (Presenter)

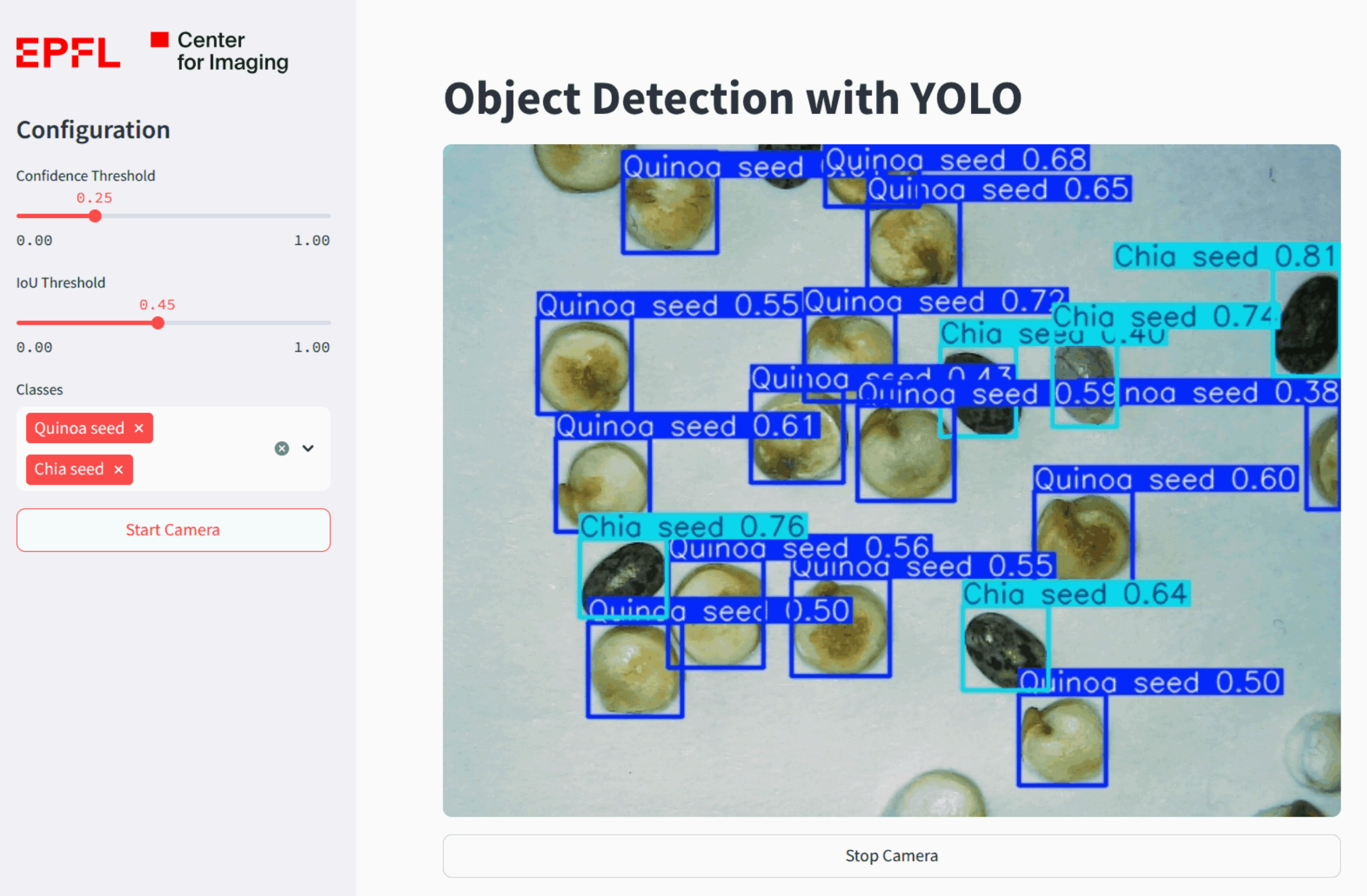
Florian Aymanns

Jules Vanaret

Edward Andò

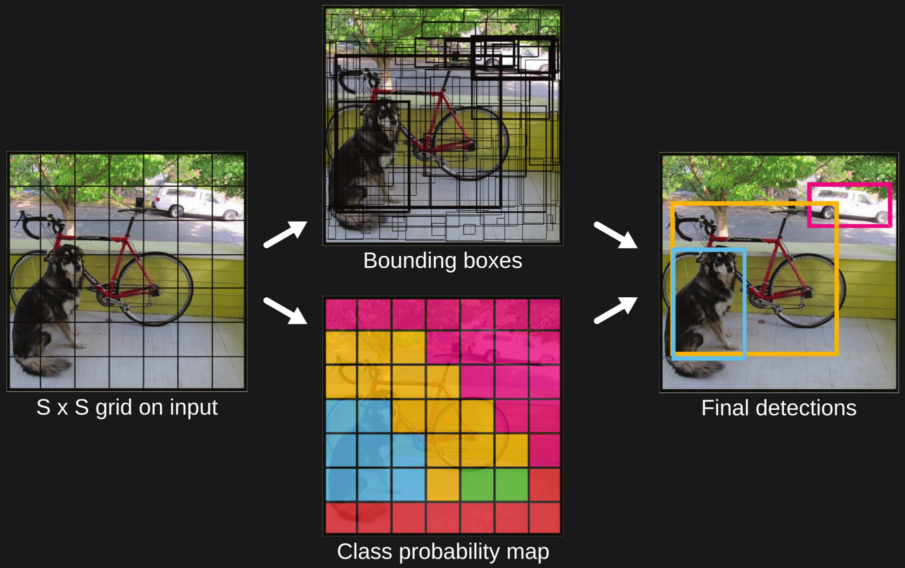
Daniel Sage

LET'S BUILD A SMART MICROSCOPE

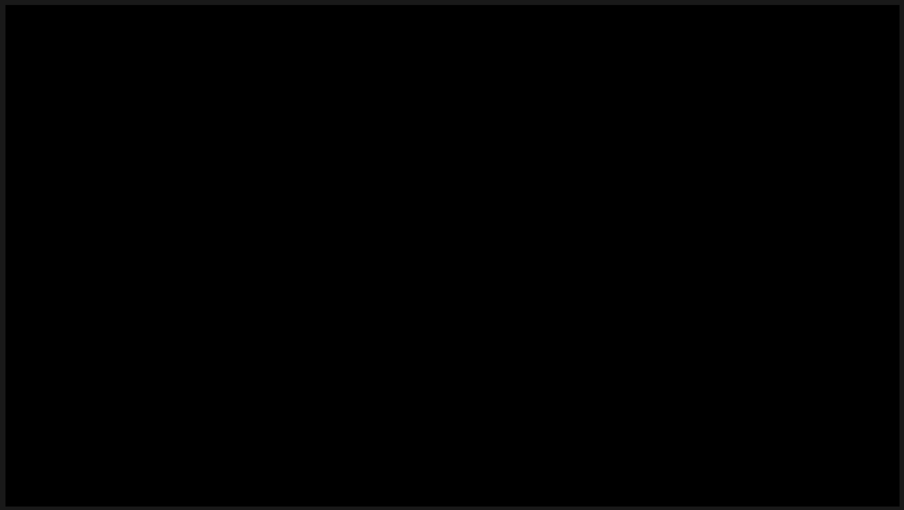


YOLO (YOU ONLY LOOK ONCE)

Detect bounding boxes and **classify** them in one go.



Original YOLO detector (2015)



YOLO11 (2025)

Source: Ultralytics

YOLO (YOU ONLY LOOK ONCE)

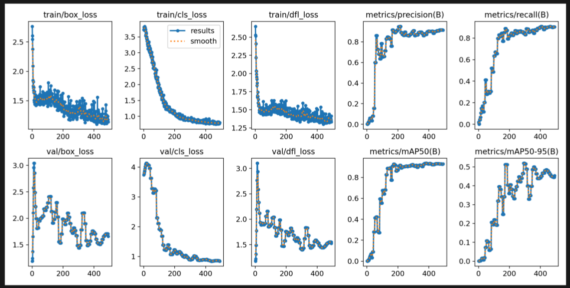
- Optimized for **real-time applications**.
- We'll use the implementation from the Ultralytics library in Python.
  - Simple, well-maintained, consistent API (*train*, *predict*, etc.).
  - Provides **pretrained models**, **data augmentations**, and **sensible defaults**.
  - Licensed under AGPL-3.0.

YOUR IMAGING EQUIPMENT FOR TODAY

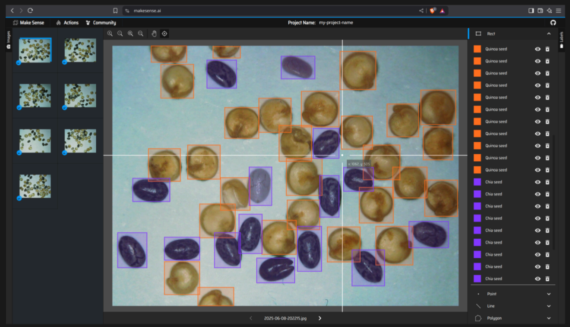
USB microscopes

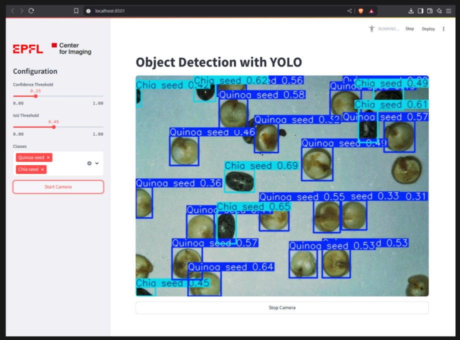
Magnification:	60-200 X
Connection:	USB-A
Cost:	~70 CHF

OVERVIEW OF THE STEPS INVOLVED



🔬 Acquisition → 🖌️ Annotation → 🎓 Training → ⚖️ Validation → 🔋 Inference





WHAT WE HOPE YOU'LL TAKE AWAY WITH YOU

- ❌ Not the microscopes; please return them at the end of the session! :)
- ✅ Understand the main steps of a computer vision project through practice.
- ✅ Get new ideas and insights that you can apply to your own projects.

YOUR TURN

🕒 UNTIL 15:30.

Make groups of 3

🧑‍🔬 👩🏻‍🔬 👩🏾‍🔬

You should be equipped with:

- 🔬 1X USB microscope
- 💻 Laptop with Python + USB-A
- 🫘 Some dry seeds

⬇️ Workshop guide

https://go.epfl.ch/yolo-workshop